

Problem Statement

Evaluate the limit from the image "dailyintegral-limit-limit-1x1-2026-08-04.jpg":

$$\lim_{n \rightarrow \infty} \sqrt{n} x_n$$

given that $x_1 \in (0, 1)$ and the sequence is defined by $x_{n+1} = \sin(x_n)$ for $n \geq 1$.

Detailed Solution

To evaluate the limit, we will first determine the convergence of the sequence (x_n) , and then apply the Stolz-Cesàro theorem to find the limit of the scaled sequence.

Step 1: Convergence of x_n

We are given $x_1 \in (0, 1)$. For $x \in (0, 1)$, we know that $0 < \sin(x) < x$. Therefore, $0 < x_2 = \sin(x_1) < x_1$. By induction, the sequence (x_n) is strictly decreasing and bounded below by 0:

$$0 < x_{n+1} < x_n < 1$$

By the Monotone Convergence Theorem, the sequence (x_n) must converge to some limit L . Taking the limit of both sides of the recurrence relation $x_{n+1} = \sin(x_n)$ gives $L = \sin(L)$. The only solution to this equation in the real numbers is $L = 0$. Thus:

$$\lim_{n \rightarrow \infty} x_n = 0$$

Step 2: Apply the Stolz-Cesàro Theorem

We want to find $\lim_{n \rightarrow \infty} \sqrt{n} x_n$. Let's examine the squared limit, $L^2 = \lim_{n \rightarrow \infty} n x_n^2$. We can rewrite this limit as:

$$\lim_{n \rightarrow \infty} \frac{n}{\frac{1}{x_n^2}}$$

Since $\lim_{n \rightarrow \infty} \frac{1}{x_n^2} = \infty$ and the sequence $1/x_n^2$ is strictly increasing, we can apply the Stolz-Cesàro theorem. This theorem states that the limit is equal to:

$$\lim_{n \rightarrow \infty} \frac{(n+1) - n}{\frac{1}{x_{n+1}^2} - \frac{1}{x_n^2}} = \lim_{n \rightarrow \infty} \frac{1}{\frac{1}{\sin^2(x_n)} - \frac{1}{x_n^2}}$$

Step 3: Evaluate the limit using Taylor series

Since $x_n \rightarrow 0$ as $n \rightarrow \infty$, we can evaluate the limit in the denominator by taking the continuous limit as $x \rightarrow 0$:

$$\lim_{x \rightarrow 0} \left(\frac{1}{\sin^2(x)} - \frac{1}{x^2} \right) = \lim_{x \rightarrow 0} \frac{x^2 - \sin^2(x)}{x^2 \sin^2(x)}$$

We use the Taylor series expansion for $\sin(x)$ centered at $x = 0$:

$$\sin(x) = x - \frac{x^3}{6} + \mathcal{O}(x^5)$$

Squaring this expression yields:

$$\sin^2(x) = \left(x - \frac{x^3}{6} + \mathcal{O}(x^5)\right)^2 = x^2 - \frac{x^4}{3} + \mathcal{O}(x^6)$$

Substituting this back into the numerator:

$$x^2 - \sin^2(x) = x^2 - \left(x^2 - \frac{x^4}{3} + \mathcal{O}(x^6)\right) = \frac{x^4}{3} - \mathcal{O}(x^6)$$

For the denominator, we have:

$$x^2 \sin^2(x) = x^2(x^2 + \mathcal{O}(x^4)) = x^4 + \mathcal{O}(x^6)$$

Now, the continuous limit becomes:

$$\lim_{x \rightarrow 0} \frac{\frac{x^4}{3} - \mathcal{O}(x^6)}{x^4 + \mathcal{O}(x^6)} = \lim_{x \rightarrow 0} \frac{\frac{1}{3} - \mathcal{O}(x^2)}{1 + \mathcal{O}(x^2)} = \frac{1}{3}$$

Therefore, the difference in the denominator of our Stolz-Cesàro application converges to $1/3$.

$$\lim_{n \rightarrow \infty} \frac{1}{\frac{1}{\sin^2(x_n)} - \frac{1}{x_n^2}} = \frac{1}{1/3} = 3$$

This means $\lim_{n \rightarrow \infty} nx_n^2 = 3$.

Step 4: Final Calculation

Since we established that $x_n > 0$ for all n , we can safely take the principal square root to find the limit of the original expression:

$$\lim_{n \rightarrow \infty} \sqrt{nx_n} = \sqrt{\lim_{n \rightarrow \infty} nx_n^2} = \sqrt{3}$$

Final Answer:

$$\sqrt{3}$$